

# **Project Overwatch**

## **Build Specification & Engineering Plan**

Unified AML Transaction Monitoring Platform  
Consolidated Fiat & Regulated Stablecoin Transactions

AnchorPoint (SCB x HKT x Animoca Brands JV) | USDC  
Hong Kong Payments / Remittance Platform

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# 1. Build Overview & Scope

## 1.1 Purpose

This Build Specification translates the Project Overwatch requirements into an actionable engineering blueprint. It defines system architecture, module boundaries, data contracts, infrastructure topology, CI/CD pipelines, testing strategy, phased delivery plan, and team sizing required to build the unified AML transaction monitoring platform. Audience: Engineering leads, architects, platform engineers, QA leads, and delivery managers.

## 1.2 What We Are Building

A unified, real-time AML transaction monitoring platform applying consistent compliance controls across fiat payment rails and regulated stablecoin (AnchorPoint HKDR and USDC) on-chain transfers. Serves a Hong Kong-based payments/remittance platform distributing AnchorPoint (SCB x HKT x Animoca Brands JV) and USDC across EVM chains and Solana.

## 1.3 Scope Boundaries

| In Scope                                                                                                   | Out of Scope (Future / Adjacent)                                    |
|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Transaction Monitoring Engine with configurable rules                                                      | KYC/KYB identity verification (consumed from existing provider)     |
| Real-time sanctions screening (OFAC, UN, EU, HK, custom lists)                                             | Customer-facing onboarding portal or mobile app                     |
| Blockchain indexer for EVM + Solana (AnchorPoint + USDC)                                                   | Core banking / ledger transaction processing                        |
| Wallet screening via blockchain analytics vendor API                                                       | Smart-contract-level compliance controls (Phase 5 R&D)              |
| Travel Rule messaging via TRP protocol                                                                     | Full graph analytics and entity relationship detection (Phase 4)    |
| Case management system with alert investigation workflow                                                   | Direct integration with all global regulatory filing portals        |
| AML Admin Portal (KPI dashboard, case management, network graph)                                           | General ledger or accounting integration                            |
| Regulatory reporting (SAR filing, HKMA returns)                                                            | Multi-jurisdiction licensing across EU / US (documented for future) |
| On-prem / hybrid infrastructure with data residency controls                                               | Public cloud-only deployment                                        |
| Audit trail + tamper-proof forensic audit ledger                                                           | Real-time screening for off-platform P2P transfers                  |
| Unified data model spanning fiat + stablecoin (27 entities: party, account, transaction, alert, case, STR, | External CRM or ERP integration                                     |

|                                                                                                                                                                                                 |                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| graph, batch, KPI)                                                                                                                                                                              |                                        |
| T+1 batch scanning: behavioral baselining, peer group comparison, multi-account structuring, network graph anomaly detection, ensemble ML scoring, cross-rail layering, sanctions retrospective | Real-time graph exploration (Phase 4+) |
| Graph analytics: PageRank, Louvain community detection, cycle detection, shortest path (Apache AGE)                                                                                             | Full-featured graph database (Neo4j)   |

## 1.4 Key Design Decisions

| Decision               | Choice                          | Rationale                                                                       |
|------------------------|---------------------------------|---------------------------------------------------------------------------------|
| Core engine            | Build (Python + FastAPI)        | Differentiation on rule flexibility, ML scoring, tight HKMA integration         |
| Event backbone         | Apache Kafka (on-prem)          | Proven at scale; replay, partitioning, exactly-once                             |
| Workflow orchestration | Temporal.io (on-prem)           | Durable execution for Travel Rule + case state machines                         |
| Database (primary)     | PostgreSQL 16+                  | ACID, JSONB, strong encryption, audit capabilities                              |
| Forensic DB            | PostgreSQL 16+ (WORM, separate) | Cryptographically sealed, independent verifiability for all manual interactions |
| Graph layer            | Apache AGE (PG extension)       | openCypher + relational SQL; no separate graph DB                               |
| Scheduling             | Apache Airflow                  | Industry standard; Kafka, PostgreSQL, cloud integration                         |
| Orchestration          | Kubernetes (k3s/Rancher)        | Portability, auto-scaling, standardised deployment                              |
| IAM                    | Keycloak (self-hosted)          | RBAC, OIDC federation, segregation of duties                                    |
| Blockchain analytics   | Integrate (vendor adapter)      | TRM Labs or Chainalysis; switching via abstracted adapter                       |
| Travel Rule            | Notabene TRP                    | Largest counterparty directory; HK VASP adoption                                |
| Frontend               | Next.js + React + TypeScript    | Established stack; SSG for dashboard; SSR for case mgmt                         |

## 2. System Context & Interfaces

### 2.1 External System Context

| External System                | Type     | Interface                            | Direction     | Protocol                   |
|--------------------------------|----------|--------------------------------------|---------------|----------------------------|
| Payments / Remittance Platform | Upstream | Sync hold/release + async events     | Bidirectional | REST + Kafka               |
| Banking Rails (SWIFT/FPS/ACH)  | Upstream | Bank file feeds                      | Inbound       | ISO 20022 / SWIFT MT / CSV |
| KYC / KYB Provider             | Upstream | Webhook + REST pull                  | Inbound       | REST (OAuth2)              |
| Blockchain Networks            | Upstream | RPC + events                         | Inbound       | JSON-RPC / WebSocket       |
| Blockchain Analytics Vendor    | External | Wallet screening API                 | Outbound      | REST (API key)             |
| Sanctions List Provider        | External | SFTP feed + REST                     | Inbound       | SFTP / HTTPS               |
| Travel Rule Counterparties     | External | TRP messaging                        | Bidirectional | HTTPS mTLS                 |
| JFIU (Hong Kong FIU)           | External | STR filing                           | Outbound      | STREAMS 2 / SFTP XML       |
| Compliance Team                | User     | Web UI                               | Bidirectional | HTTPS                      |
| Internal Audit / Regulators    | User     | Read-only portal + forensic verifier | Inbound       | HTTPS + SFTP               |

### 2.2 Deployment Context

The platform deploys on dedicated on-premises Kubernetes clusters (Hong Kong data centre: Equinix HK, MEGA-i). All customer PII and transaction data resides on-prem. Vendor services receive only pseudonymised data (wallet hashes). Travel Rule data is encrypted end-to-end.

### 3. Module Decomposition & Responsibilities

The system decomposes into eleven microservices. Services communicate via Kafka events (async) for most interactions, with REST/gRPC for synchronous request-response. Each owns a distinct bounded context.

#### 3.1 Ingestion Layer

##### 3.1.1 ingestion-service

- Language: Python 3.12+ (FastAPI + Kafka producer)
- Fiat: POST /api/v1/transactions/fiat (sync hold/release); batch SFTP for SWIFT/ISO 20022
- On-chain: RPC poller per EVM chain; Solana geyser; WebSocket receiver
- Normalisation: transforms to Canonical Transaction Schema (Section 4.2)
- Enrichment: wallet screening cache for address risk scores pre-publish

##### 3.1.2 Kafka Topic Map

| Topic               | Partitions | Retention | Key                | Consumers                                          |
|---------------------|------------|-----------|--------------------|----------------------------------------------------|
| fiat.raw            | 6          | 7d        | transaction_ref    | normalisation engine                               |
| onchain.raw         | 12         | 7d        | chain_id+tx_hash   | normalisation engine                               |
| fiat.normalised     | 6          | 30d       | customer_id        | TME, sanctions, risk scoring                       |
| onchain.normalised  | 12         | 30d       | customer_id/wallet | TME, sanctions, blockchain analytics               |
| screening.requests  | 6          | 7d        | round-robin        | sanctions, wallet screening                        |
| screening.results   | 6          | 30d       | transaction_ref    | TME, audit                                         |
| travelrule.requests | 3          | 30d       | round-robin        | Travel Rule gateway                                |
| travelrule.results  | 3          | 90d       | transaction_ref    | TME, audit                                         |
| alerts.generated    | 6          | 90d       | customer_id        | CMS, audit                                         |
| audit.events        | 3          | permanent | event_id           | audit service                                      |
| forensic.events     | 6          | permanent | event_category     | forensic-audit-service (sole writer to ForensicDB) |

#### 3.2 Transaction Monitoring Engine (TME)

##### tme-engine

- Language: Python 3.12+ (FastAPI + Kafka consumer)

- Internal: in-memory sliding window per customer (5min, 1hr, 24hr)
- Rules: YAML DSL; versioned in PostgreSQL; hot-reload
- Pipeline: Pre-filter > Tier gate > Rule evaluation (parallel async) > ML scoring > Composite score > Decision > Output
- Decisions: score 0-30 auto-approve; 31-65 flag for review; 66-100 block

#### tme-ml (ML Inference)

- ONNX runtime for XGBoost/LightGBM
- Features: amount log, frequency, z-score baseline, wallet risk, counterparty risk, jurisdiction pair, time-of-day
- Output: probability [0,1] + SHAP explanations

### 3.3 Sanctions Screening Service

- Python 3.12+ (FastAPI sync + Kafka batch)
- Fuzzy name matching: rapidfuzz (Levenshtein, token-set ratio, Soundex)
- Wallet screening: in-memory hash map for sub-ms exact-match lookups
- Lists: OFAC SDN, UN, EU, HKMA, custom; daily SFTP pull; diff detection
- Confidence:  $\geq 80$  = block; 60-79 = review
- Batch re-screening: all customers within 24h of list update

### 3.4 Blockchain Indexer & Wallet Analytics Adapter

#### blockchain-indexer

- Language: Go (performance-critical polling)
- Per-chain goroutine worker; monitors Transfer events for AnchorPoint + USDC
- Chains: Ethereum, Polygon, Arbitrum, Optimism, Base, Solana
- Publishes to onchain.raw (JSON: chain\_id, tx\_hash, block, from, to, value, token, timestamp)
- Writes stablecoin\_txn\_detail records extending the canonical transaction with on-chain metadata

#### wallet-analytics-adapter

- Python 3.12+; abstracted vendor interface (TRM / Chainalysis / Elliptic)
- Redis cache: 24h TTL static scores, 1h TTL dynamic scores
- N-hop chain analysis (configurable 1-3 hops); circuit breaker >5 failures/min
- Caches wallet risk scores used to populate stablecoin\_txn\_detail.ba\_risk\_score and enable account.account\_risk\_rating updates

### 3.5 Travel Rule Gateway

#### travelrule-gateway

- Python 3.12+ (Temporal workflow worker); triggered  $\geq$  HKD 8,000 equivalent
- Workflow: Counterparty discovery (Notabene) > TRP message > await response > timeout (5min) > hold/unhosted EDD
- Publishes result to travelrule.results; TME consumes for final decision

### 3.6 Risk Scoring Service

- Python 3.12+ (FastAPI + PostgreSQL)
- Factors: velocity deviation, wallet risk, counterparty risk, sanctions, PEP, adverse media, jurisdiction, source-of-funds
- Triggers: new txn, sanctions update, PEP update, manual re-score

### 3.7 Case Management System (CMS)

#### case-management-service

- Python 3.12+ (FastAPI + Temporal state machines)
- States: NEW > ASSIGNED > UNDER\_INVESTIGATION > CLOSED\_FP / CLOSED\_MONITORING / SAR\_FILED
- Temporal-enforced transitions; no direct DB mutation

| Workflow         | Trigger                   | Steps                                                           | SLA         |
|------------------|---------------------------|-----------------------------------------------------------------|-------------|
| Alert assignment | New alert                 | Assign (round-robin); notify; escalate if unassigned >1hr       | Seconds     |
| Investigation    | Analyst self-assigns      | Hold transaction; gather evidence; blockchain queries; document | 4hr typical |
| SAR approval     | Analyst drafts SAR        | Peer review > manager > compliance officer > file to JFIU       | 24hr        |
| Escalation       | High-risk unactioned >1hr | Team lead > compliance manager > MLRO                           | Progressive |

### 3.8 AML Admin Portal

- Framework: Next.js 14+ (App Router) + React + TypeScript
- UI: shadcn/ui (Radix + Tailwind); state: React Query + Zustand; viz: Cytoscape.js + Recharts

| Route        | Module          | Description                                                   |
|--------------|-----------------|---------------------------------------------------------------|
| /dashboard   | KPI Dashboard   | Real-time AML metrics; configurable date range; exportable    |
| /alerts      | Alert Queue     | Paginated, filterable, sortable alert list                    |
| /alerts/[id] | Alert Detail    | Triggered rules, risk breakdown, customer 360, wallet summary |
| /cases       | Case Management | Case list: status, priority, assignee;                        |

|                    |                    |                                                                             |
|--------------------|--------------------|-----------------------------------------------------------------------------|
|                    |                    | bulk assignment                                                             |
| /cases/[id]        | Case Detail        | Investigation workspace: timeline, graph, notes, evidence, SAR draft        |
| /reports           | Reporting          | SAR generation, HKMA returns, scheduled reports                             |
| /batch-scan        | T+1 Batch Scanning | Batch run history, daily briefing reports, ad-hoc trigger, analysis results |
| /config/rules      | Rule Management    | Rule config, version history, enable/disable, shadow-mode                   |
| /config/thresholds | Thresholds         | Per-tier and per-jurisdiction thresholds                                    |
| /admin/users       | User Management    | Keycloak-mediated provisioning and role assignment                          |
| /audit/logs        | Audit Log Viewer   | Immutable audit log search, filter, export                                  |
| /forensic/verify   | Forensic Verifier  | Independent hash-chain verification UI for auditors                         |

### 3.9 Regulatory Reporting & SAR Filing

- Python 3.12+; SAR: JFIU-compatible STR form XML from case data
- Filing: STREAMS 2 web portal (assisted upload); SFTP XML (future automated)
- HKMA returns: automated periodic AML statistical returns via Airflow DAGs

### 3.10 Audit Trail Service

- Python 3.12+ (idempotent Kafka consumer)
- Consumes audit.events topic; stores in dedicated PostgreSQL schema with monthly partitioning
- Append-only (no UPDATE/DELETE); hash-chain: SHA-256(prev\_hash + payload)
- Integrity checker runs hourly; events: transaction decisions, rule evaluations, screening results, system events

### 3.11 Tamper-Proof Forensic Audit Ledger (ForensicDB)

The ForensicDB is a dedicated, cryptographically-sealed write-once-read-many (WORM) database capturing every manual human interaction: compliance analyst decisions, SAR approvals, rule adjustments, risk tier overrides, and configuration changes. It provides forensic-grade traceability for regulatory examination, internal audit, and legal discovery, with independently verifiable proof that no record has been altered or deleted after the fact.

### 3.11.1 Design Principles

- WORM immutability: records cannot be modified, deleted, or soft-deleted; enforced at storage engine level, not application logic
- Cryptographic sealing: every record SHA-256 hashed and chained; tampering breaks the chain and is independently detectable
- Operator attribution: all manual actions signed with operator Keycloak identity, enriched with session token metadata, and recorded with hardware-level NTP-synchronised timestamp
- Segregation: ForensicDB runs on physically separate PostgreSQL instance from operational DB; separate replication topology, access controls, and backup schedule
- Independent verifiability: standalone verifier tool (CLI + web UI) recomputes the entire hash chain without any application code; suitable for external auditors and regulators
- Retention: minimum 7 years (AMLO requirement); automatic annual archival to WORM optical media with chain-of-custody tracking

### 3.11.2 What Gets Recorded (Forensic Event Categories)

| Event Category            | Examples                                          | Recorded Artifacts                                                                                         |
|---------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Alert disposition         | Analyst closes alert as false positive            | Alert ID, score, triggered rules, pre-state, decision, justification, operator ID, session hash, timestamp |
| Case decisions            | Manager approves SAR filing; analyst escalates    | Case ID, state transition (from->to), notes, evidence hashes, approval chain, SAR draft version hash       |
| Rule configuration        | Compliance manager adjusts velocity threshold     | Rule ID, old/new definition (full JSON), diff, change ticket ref, approver, effective timestamp            |
| Risk tier overrides       | Analyst downgrades customer from HIGH to MEDIUM   | Party ID, old/new tier, override reason (free-text + category), expiry, approver, supporting doc hashes    |
| Sanctions list overrides  | Analyst marks sanctions hit as false positive     | Screening result ID, matched entity, match confidence, override reason, evidence refs, dual-approval chain |
| Travel Rule interventions | Compliance officer releases held transaction      | Transaction ID, workflow ID, hold reason, release reason, counter signatory, counterparty comms logs       |
| User access events        | Analyst views sensitive customer; auditor exports | User ID, session ID, resource, access type (read/write/export),                                            |

|                              |                                               |                                                                                                           |
|------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------|
|                              |                                               | client IP, user agent, timestamp                                                                          |
| <b>System config changes</b> | DevOps adjusts Kafka partition count          | Config key, old/new value hash, change management ticket, approver, deployment timestamp                  |
| <b>Model governance</b>      | ML model promoted from shadow to active       | Model ID, version, training run, validation metrics, approval chain, effective timestamp, rollback policy |
| <b>SAR/STR lifecycle</b>     | SAR draft, review, approval, filing, JFIU ack | Full SAR XML at each stage, each reviewer ID + timestamp, JFIU ref, filing timestamp                      |

### 3.11.3 Cryptographic Sealing Architecture

The ForensicDB employs dual-layer sealing: per-table Merkle trees for efficient partial verification and a global hash chain for end-to-end integrity.

- Record-level: SHA-256(record\_type, payload, operator\_id, business\_timestamp, previous\_record\_hash) computed at write time
- Merkle tree (per table): balanced Merkle tree per event category. Root published hourly to external witness. Auditors verify single records in  $O(\log n)$  time.
- Global hash chain: chain\_n = SHA-256(chain\_n-1, table\_name, record\_hash, timestamp). Monotonic sequence number. Chain head published with Merkle roots.
- Verification tool (forensic-verifier CLI): recomputes all hashes from genesis; outputs tamper report: clean or specific divergent records
- Auto-verification: integrity checker runs every 15 minutes on read-only replica; chain break triggers immediate PO alert

### 3.11.4 WORM Enforcement at the Storage Layer

- Database-level: forensic\_db role has only INSERT + SELECT. UPDATE, DELETE, TRUNCATE, DROP revoked. Table ownership held by separate admin role unused by application.
- Row-level: BEFORE UPDATE / BEFORE DELETE triggers unconditionally raise exceptions. Trigger removal itself generates an audit event in externally-monitored tamper log.
- Filesystem-level: PostgreSQL data directory mounted with chattr +i on WAL directory. Underlying block device exported from storage array with WORM snapshot policies.
- Replication: read-only replica in physically separate network segment serves verifier and auditor access. No write path to replica.
- Backup: daily encrypted backups to MinIO with S3 Object Lock in Compliance mode (retention lock, no deletion even by root). Annual archival to WORM optical media with chain-of-custody.

### 3.11.5 External Witness Publishing

To provide independent third-party verifiability, Merkle roots and chain heads are published to external witnesses.

- Hourly: composite hash (Merkle roots + chain head) published to Bitcoin testnet via OP\_RETURN or to Polygon (lower cost). Transaction ID stored in ForensicDB as external witness reference.
- Daily attestation: signed attestation containing global chain head and all Merkle roots stored in separate cloud immutable store (S3 Object Lock / Azure Immutable Blob) with different access control boundary.
- Independent verification: any auditor can independently verify by: (a) recomputing the hash chain from ForensicDB records; (b) verifying published Merkle root matches recomputed root; (c) checking blockchain witness transaction exists and contains the same hash. No trust in the institution systems is required.

### 3.11.6 Service: forensic-audit-service

- Language: Python 3.12+ (idempotent Kafka consumer + REST API for auditors)
- Writes: consumes forensic.events Kafka topic (partitioned by event\_category); inserts into ForensicDB as sole writer
- Reads: REST API - GET /forensic/records, GET /forensic/chain/verify (returns current chain state and last verified timestamp)
- Verifier: scheduled subprocess recomputes hash chains in read-only mode against replica; results logged to Prometheus metrics and tamper-alert topic
- Witness publisher: computes current Merkle roots, publishes to external witness, stores witness tx reference
- Retention manager: exports records older than 7 years to WORM media, verifies export hash, logs archival with chain-of-custody tracking

### 3.11.7 ForensicDB Schema

| Table                             | Key Columns                                                                                                                          | Purpose                                                  |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>forensic.chain_head</b>        | sequence_id (bigserial), chain_hash, record_table, record_hash, business_timestamp, witness_tx_id                                    | Global hash chain linking all records in insertion order |
| <b>forensic.alert_disposition</b> | event_id (UUID v7), alert_id, disposition, justification, operator_id, pre_state (jsonb), post_state (jsonb), prev_hash, record_hash | Every alert decision with full before/after state        |
| <b>forensic.case_transition</b>   | event_id, case_id, from_state, to_state, operator_id, notes, evidence_hashes[], approval_chain (jsonb), prev_hash, record_hash       | Case state transitions with approval chain               |
| <b>forensic.rule_change</b>       | event_id, rule_id, old_definition (jsonb), new_definition (jsonb), change_diff, approver_id, change_ticket_ref, prev_hash,           | All rule and threshold configuration changes             |

|                                  |                                                                                                                                                 |                                                                    |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|                                  | record_hash                                                                                                                                     |                                                                    |
| <b>forensic.risk_override</b>    | event_id, party_id, old_tier, new_tier, override_reason, override_category, expiry_dt, approver_id, supporting_docs[], prev_hash, record_hash   | Manual risk tier overrides with expiry and justification           |
| <b>forensic.sar_lifecycle</b>    | event_id, str_id, stage (DRAFT/REVIEWED/APPROVED/FILED/ACKED), operator_id, sar_content (xml), reviewer_ids[], jfiu_ref, prev_hash, record_hash | Complete SAR lifecycle with full XML at each stage                 |
| <b>forensic.access_log</b>       | event_id, user_id, session_id, resource_type, resource_id, access_type, client_ip, user_agent, justification, prev_hash, record_hash            | All access to sensitive customer data and case records             |
| <b>forensic.merkle_root</b>      | table_name, block_start, block_end, merkle_root, computed_at, witness_tx_id                                                                     | Published Merkle roots per table per hour for partial verification |
| <b>forensic.external_witness</b> | witness_id, composite_hash, target (BITCOIN_TESTNET/POLYGON/S3), witness_tx_id, published_at, verified_at                                       | Record of all external witness publications                        |

### 3.11.8 Integration with Existing Services

The ForensicDB complements, not replaces, the operational audit trail (audit.events / aml\_audit). The operational audit trail captures system-level events for observability and debugging. The ForensicDB captures manual human interactions for regulatory-grade tamper-proof evidence. They operate on physically separate infrastructure.

- TME: publishes rule execution logs to audit.events (operational) only; forensic events generated only if compliance analyst manually overrides a TME decision
- CMS: publishes all alert dispositions, case state transitions, notes, evidence uploads, and SAR drafts to forensic.events topic
- AML Admin Portal: publishes all rule config changes, risk overrides, threshold adjustments, and access to sensitive records to forensic.events
- regulatory-reporting-service: publishes SAR lifecycle events to forensic.events
- All forensic.events publishers must include: operator\_id (Keycloak), session\_id, client\_ip, user\_agent, business justification. Missing fields = rejected by forensic-audit-service.
- forensic-audit-service is the sole writer to the ForensicDB. No other service has credentials to the ForensicDB instance. The operational audit trail continues to write to the separate aml\_audit PostgreSQL instance.



### 3.12 T+1 Batch Transaction Scanning (batch-scanning-service)

The T+1 Batch Scanning system provides a complementary, deeper analytical layer to the real-time transaction monitoring engine. While the real-time TME processes each transaction individually within sub-500ms latency using in-memory sliding windows, the T+1 batch scanner processes the complete previous day's transaction set in bulk, enabling computationally intensive analyses that are infeasible in the real-time path: full customer behavioral baselining, peer-group comparative analysis, multi-day pattern detection, ML model scoring on complete transaction sets, and retrospective rule evaluation with adjusted parameters. The batch scanner references the `detection_scenario` table — scenarios marked `processing_mode=BATCH` or `processing_mode=BOTH` are eligible for batch execution. The `batch_job` table tracks each scheduled and ad-hoc run. The `kpi_snapshot` and `scenario_kpi` tables are populated by the batch pipeline for regulatory reporting and rule effectiveness monitoring.

#### 3.12.1 Design Rationale

Real-time AML monitoring necessarily trades depth for speed. A real-time engine must render a decision in milliseconds using only the data available at that instant — limited to sliding-window velocity counters and point-in-time sanctions checks. T+1 batch scanning removes the latency constraint, allowing the system to: (a) compare each transaction against the customer's complete daily profile rather than a 24-hour sliding window; (b) detect patterns that only emerge across an entire business day (e.g., coordinated structuring across multiple accounts); (c) run computationally expensive ML models (full feature vectors, ensemble models, graph-based anomaly detection) that cannot execute in real-time; (d) apply updated rule thresholds retrospectively to catch transactions that would have been flagged under newly-tuned parameters; and (e) generate enriched risk intelligence (peer group deviation scores, behavioral trend lines, network centrality metrics) for compliance analyst morning briefings.

- Separation of concerns: The real-time TME is optimised for speed and availability (p95 <500ms, 99.95% uptime). The batch scanner is optimised for analytical depth and completeness, running on a relaxed schedule (T+1 morning window) with significantly higher resource allocation per transaction.
- Idempotency: Batch scanner marks each processed transaction with a `batch_scan_id` to prevent double-counting. Re-runs for the same date overwrite previous batch results with updated analysis timestamps.
- Complementarity: Batch scanner does NOT duplicate real-time rules. It augments with analyses the real-time engine cannot perform. Flagged batch anomalies create separate "T+1 alerts" that are lower urgency than real-time alerts but feed into the same case management workflow for investigation.

#### 3.12.2 Architecture & Data Flow

The batch-scanning-service is implemented as a Python 3.12+ service orchestrated by Apache Airflow DAGs. It runs daily at 02:00 HKT (after the previous day's transactions have fully settled and all blockchain finality confirmations are complete).

| Stage              | Component      | Description       | Data Source        | Output         |
|--------------------|----------------|-------------------|--------------------|----------------|
| 1. Data Collection | Airflow Sensor | Wait for previous | Prometheus metrics | Proceed signal |

|                   |                        |                                                                                                                                                                                                                                                                                                       |                                                          |                                              |
|-------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------|
|                   |                        | day's data to be fully ingested and blockchain finality confirmed (all chains). Polls ingestion-service health + blockchain-indexer lag metrics.                                                                                                                                                      |                                                          |                                              |
| <b>2. Extract</b> | batch-scanning-service | Pull all canonical_transaction records for the target date (00:00:00 to 23:59:59 HKT) from aml_core. Includes all real-time decisions, scores, and rule triggers for comparison.                                                                                                                      | aml_core (canonical_transaction MV)                      | Daily transaction dataset (Parquet in MinIO) |
| <b>3. Enrich</b>  | batch-scanning-service | Augment each transaction with: full customer profile (current risk tier, PEP status, account age, declared income), wallet analytics scores (refreshed from vendor API if cache expired), peer group assignment (by segment, jurisdiction, product usage), and real-time TME decision for comparison. | aml_core, wallet-analytics-adapter, risk-scoring-service | Enriched dataset                             |
| <b>4. Analyse</b> | batch-scanning-service | Execute batch-specific analyses (see 3.12.3). Each analysis runs as a parallel worker in a Kubernetes Job with configurable resource limits.                                                                                                                                                          | Enriched dataset + historical baselines (TimescaleDB)    | Analysis results                             |

|                    |                        |                                                                                                                                                                                                               |                                      |                                                        |
|--------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------|
| <b>5. Score</b>    | batch-scanning-service | Apply batch ML model (full-feature XGBoost ensemble with graph features) to the complete daily dataset. Generate risk scores with SHAP explanations.                                                          | ML model from MinIO                  | Batch risk scores                                      |
| <b>6. Alert</b>    | batch-scanning-service | Generate T+1 alerts for anomalous findings. Publish to Kafka topic batch.alerts with lower priority than real-time alerts. Auto-create cases in CMS for high-confidence findings.                             | Analysis results + batch risk scores | batch.alerts topic + CMS cases                         |
| <b>7. Report</b>   | batch-scanning-service | Generate Daily AML Briefing report: top 10 riskiest transactions, peer group anomaly summary, rule effectiveness metrics (real-time vs batch detection), false positive trend, new pattern detection summary. | All batch results                    | Daily Briefing PDF in MinIO + email to compliance team |
| <b>8. Feedback</b> | batch-scanning-service | Compare real-time TME decisions against batch analysis findings. Log discrepancies as feedback for rule tuning. Update ML training dataset with batch labels.                                                 | Real-time vs batch comparison        | Feedback to rule config + ML training pipeline         |

### 3.12.3 Batch-Only Analytical Capabilities

The following analyses execute exclusively in the batch path because they require the complete daily transaction set, historical baselines, or computationally expensive models:

| Analysis                                           | Description                                                                                                                                                                                                                                                                                        | Data Required                                                   | Output                                                                        |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Full-Day Customer Behavioral Baselineing</b>    | Compute complete daily statistics per customer: total volume, txn count, average amount, beneficiary diversity, time-of-day distribution, rail mix (fiat vs stablecoin ratio). Compare against 30/90-day historical baseline using statistical deviation tests (z-score, MAD).                     | All customer txns for the day + 90-day historical (TimescaleDB) | Behavioral deviation score per customer; anomaly flags for >3 sigma deviation |
| <b>Peer Group Comparative Analysis</b>             | Assign each customer to a peer group (by segment, jurisdiction, declared income band, product usage). Compute group-level statistics (median daily volume, interquartile ranges). Flag customers whose daily activity deviates significantly from their peer group.                                | Customer profiles + daily txns + peer group assignments         | Peer deviation score per customer; top anomalies ranked                       |
| <b>Multi-Account Structuring Detection</b>         | Identify coordinated structuring across multiple accounts controlled by the same party or linked parties. Detect patterns where total daily outflow across linked accounts exceeds thresholds even though individual accounts stay below. Use party graph (Apache AGE) for relationship traversal. | All txns + party graph (AGE) + party_wallet linkages            | Structuring cluster alerts with linked account details                        |
| <b>Velocity Pattern Mining (Multi-Day Windows)</b> | Detect complex velocity patterns spanning multiple days: ramping (increasing amounts over 3-5 days), burst-and-pause (high activity                                                                                                                                                                | 7-day txn history per customer (TimescaleDB)                    | Pattern alerts with pattern type classification                               |

|                                               |                                                                                                                                                                                                                                                                                               |                                                                         |                                                                     |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------|
|                                               | followed by dormancy), cyclical patterns. Apply time-series decomposition and anomaly detection.                                                                                                                                                                                              |                                                                         |                                                                     |
| <b>Network Graph Anomaly Detection</b>        | Build the full daily transaction graph (nodes: parties/wallets, edges: transfers). Run graph algorithms: PageRank for influence detection, community detection for cluster identification, outlier edge detection for anomalous transfer patterns. Compare against historical graph baseline. | Full daily txn graph + historical graph snapshots (AGE)                 | Graph anomaly scores; community membership changes; new edge alerts |
| <b>Ensemble ML Scoring (Full Feature Set)</b> | Run the production ML model with the complete feature set that cannot be computed in real-time: full-day behavioral features, peer group features, graph centrality features, temporal pattern features, and cross-rail interaction features. Generate comprehensive risk scores.             | Full enriched dataset + model artifacts (MinIO)                         | ML risk score per transaction with SHAP explanation                 |
| <b>Rule Effectiveness Retrospective</b>       | For each real-time rule, compute: alert rate, false positive rate (based on analyst disposition feedback), detection overlap with batch analyses, and transactions that WOULD have been flagged under adjusted thresholds. Generate rule tuning recommendations.                              | Real-time decisions + batch results + analyst dispositions (ForensicDB) | Rule effectiveness report with tuning suggestions                   |
| <b>Cross-Rail Layering</b>                    | Full analysis of fiat-to-                                                                                                                                                                                                                                                                     | All fiat + on-chain txns for                                            | Layering alerts with full                                           |

|                                          |                                                                                                                                                                                                                               |                                    |                                                       |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------------------------------|
| <b>Detection</b>                         | stablecoin-to-fiat flows across the entire day. Detect multi-hop layering: fiat deposit > crypto purchase > transfer to unhosted wallet > transfer to exchange > fiat withdrawal. Reconstruct complete fund flow trails.      | the day (canonical_transaction)    | fund flow trail visualization data                    |
| <b>Sanctions/Screening Retrospective</b> | Re-screen all transactions against the latest sanctions list version (which may have been updated mid-day). Detect transactions that passed screening on a stale list version. Generate re-screening alerts.                  | Daily txns + latest sanctions list | Retrospective screening alerts                        |
| <b>SAR Pre-Population Intelligence</b>   | For high-risk batch findings, pre-compute the data that would be needed for a SAR: transaction timeline, counterparty details, fund flow summary, risk factor breakdown. Store as pre-populated SAR draft for analyst review. | Batch findings + case data + txns  | Pre-populated SAR drafts for high-confidence findings |

### 3.12.4 Kafka Topics

| Topic                      | Partitions | Retention | Key          | Producer               | Consumers                              |
|----------------------------|------------|-----------|--------------|------------------------|----------------------------------------|
| <b>batch.scan.commands</b> | 3          | 7d        | batch_run_id | Airflow DAG            | batch-scanning-service                 |
| <b>batch.scan.progress</b> | 3          | 7d        | batch_run_id | batch-scanning-service | Airflow DAG, monitoring                |
| <b>batch.alerts</b>        | 6          | 90d       | customer_id  | batch-scanning-service | case-management-service, audit-service |
| <b>batch.results</b>       | 6          | 365d      | batch_run_id | batch-scanning-service | aml_analytics (TimescaleDB),           |

|  |  |  |  |  |           |
|--|--|--|--|--|-----------|
|  |  |  |  |  | dashboard |
|--|--|--|--|--|-----------|

### 3.12.5 Service: batch-scanning-service

- Language: Python 3.12+ (Airflow DAGs + Kubernetes Jobs for compute-intensive analyses)
- Orchestration: Apache Airflow DAG (scheduled: 0 2 \* \* \* HKT). DAG triggers data extraction, enrichment, parallel analysis jobs, alert generation, and reporting.
- Compute: Each analysis type (3.12.3) runs as a separate Kubernetes Job with configurable resource requests. Graph analysis and ML scoring are the most resource-intensive (8 CPU, 32GB memory per job).
- Storage: Daily transaction datasets stored as Parquet files on MinIO (batch-data bucket, 90-day retention). Analysis results stored in TimescaleDB for trend analysis.
- API: POST /batch/scan/trigger (manual trigger for ad-hoc runs); GET /batch/scan/{run\_id}/status; GET /batch/scan/{run\_id}/results; GET /batch/scan/daily-briefing/{date}
- Monitoring: Prometheus metrics for batch run duration, analysis completion time per stage, alert count per run, data volume processed. Grafana dashboard: "T+1 Batch Scanning" showing run history, alert trends, and analysis performance.

### 3.12.6 Database Schema (batch-specific tables in aml\_core)

| Table                            | Key Columns                                                                                                                                               | Purpose                                                             |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>batch_scan_run</b>            | run_id (UUID), scan_date, status (PENDING/RUNNING/COMPLETED/FAILED), started_at, completed_at, txns_processed, alerts_generated, analysis_results (jsonb) | Tracks each daily batch scan execution                              |
| <b>batch_scan_result</b>         | result_id (UUID), run_id, transaction_id, analysis_type, anomaly_score, findings (jsonb), alert_generated (bool), created_at                              | Per-transaction batch analysis results, linked to batch run         |
| <b>batch_alert</b>               | alert_id (UUID), run_id, transaction_id, analysis_type, risk_score, alert_reason, priority (LOW/MEDIUM/HIGH), status, case_id (nullable), created_at      | T+1 alerts from batch scanning, lower urgency than real-time alerts |
| <b>batch_peer_group</b>          | group_id (UUID), group_name, segment, jurisdiction, income_band, product_usage, member_count, median_daily_volume, iqr_low, iqr_high, updated_at          | Peer group definitions and baseline statistics, recomputed monthly  |
| <b>batch_behavioral_baseline</b> | customer_id, baseline_date, total_volume_30d,                                                                                                             | Per-customer 30/90-day behavioral                                   |

|  |                                                                                                                         |                                   |
|--|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
|  | avg_txn_amount_30d,<br>txn_count_30d,<br>beneficiary_diversity_30d,<br>rail_mix_ratio_30d,<br>stddev_volume, updated_at | baselines for deviation detection |
|--|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------|

### 3.12.7 Integration with Existing Services

- TME: Batch scanner reads TME decisions (from canonical\_transaction) for comparison but does NOT modify real-time decisions. Rule effectiveness feedback is published as recommendations to the rule config review queue — no automatic rule changes.
- CMS: T+1 batch alerts are published to a separate batch.alerts topic with priority=LOW (vs real-time alerts which are MEDIUM/HIGH/CRITICAL). They create cases in CMS with source=BATCH\_SCAN. Analysts review batch alerts during morning briefing alongside real-time overnight alerts.
- ForensicDB: All batch-generated alerts, analyst dispositions on batch cases, and rule tuning recommendations derived from batch analysis are recorded in forensic.alert\_disposition and forensic.rule\_change tables.
- AML Admin Portal: New dashboard widgets: "T+1 Batch Findings" (daily alert count, top anomalies, peer group highlights), "Real-Time vs Batch Detection Comparison" (Venn diagram of alerts detected by each), "Rule Effectiveness" (false positive rate trends). New page: /batch-scan for viewing batch run history, results, and triggering ad-hoc runs.
- ML Pipeline: Batch scan results feed into the ML training dataset as additional labelled data. Discrepancies between real-time and batch detection serve as training signals for model improvement.
- TimescaleDB (aml\_analytics): Batch results and behavioral baselines are stored as hypertables for long-term trend analysis and regulatory reporting.

### 3.13 Real-Time vs Batch Processing: Capability Matrix

The following matrix clarifies the division of responsibility between the real-time TME and the T+1 batch scanner, ensuring no duplication of effort and clear ownership boundaries.

| Capability                              | Real-Time TME                            | T+1 Batch Scanner       | Rationale for Split                                                                                    |
|-----------------------------------------|------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Velocity checks (sliding window)</b> | Yes — 5min, 1hr, 24hr windows            | Not duplicated          | Real-time windows are sufficient for velocity; batch provides full-day baselining (different analysis) |
| <b>Threshold rules (single txn)</b>     | Yes — amount, counterparty, jurisdiction | Not duplicated          | Point-in-time rules are real-time decisions; batch retrospectively evaluates with updated thresholds   |
| <b>Sanctions screening</b>              | Yes — real-time list                     | Yes — retrospective re- | Sanctions lists update                                                                                 |

|                                            |                                      |                                                              |                                                                                                     |
|--------------------------------------------|--------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
|                                            |                                      | screen with latest list                                      | mid-day; batch catches txns screened against stale list                                             |
| <b>Full-day behavioral baselining</b>      | No — only 24hr sliding window        | Yes — complete daily profile vs 30/90-day baseline           | Requires full day of data + historical comparison; computationally expensive                        |
| <b>Peer group deviation</b>                | No                                   | Yes                                                          | Requires complete daily dataset across all customers; group statistics computed post-hoc            |
| <b>Multi-account structuring</b>           | Partial — single-account only        | Yes — cross-account, linked-party detection                  | Real-time cannot see across accounts without graph traversal; batch uses full party graph           |
| <b>Multi-day pattern mining</b>            | No                                   | Yes — ramping, burst-pause, cyclical                         | Requires complete multi-day history; computationally expensive time-series analysis                 |
| <b>Network graph anomaly detection</b>     | No                                   | Yes — PageRank, community detection, outlier edges           | Full graph construction and algorithm execution takes minutes; infeasible per-transaction           |
| <b>ML ensemble scoring (full features)</b> | No — limited features in shadow mode | Yes — full feature set including graph + peer + temporal     | Full feature computation requires batch context; ML in real-time is lightweight anomaly detection   |
| <b>Cross-rail layering (full trail)</b>    | Partial — single-hop detection       | Yes — multi-hop fund flow reconstruction                     | Requires complete daily data across rails to reconstruct full flow; real-time sees only current txn |
| <b>Rule effectiveness analysis</b>         | No                                   | Yes — false positive rates, threshold tuning recommendations | Requires analyst disposition data + full day of results; post-hoc analysis                          |
| <b>SAR pre-population intelligence</b>     | No                                   | Yes — pre-computed fund flow summaries                       | Batch can pre-compute SAR data for high-risk findings; real-time SAR is                             |

|                                        |                                 |                                     |                                                                                      |
|----------------------------------------|---------------------------------|-------------------------------------|--------------------------------------------------------------------------------------|
|                                        |                                 |                                     | analyst-initiated                                                                    |
| <b>Travel Rule</b>                     | Yes — triggered at >= HKD 8,000 | Not applicable                      | Travel Rule is a pre-settlement requirement; cannot be done T+1                      |
| <b>Immediate hold/release decision</b> | Yes — sync response required    | No — batch cannot hold transactions | Pre-settlement hold/release is a real-time function only; batch is purely analytical |

## 4. Data Architecture & Storage

The converged AML Data Model supports both real-time and T+1 batch processing, graph-based network exploration, and HKMA regulatory compliance. It spans fiat and stablecoin rails under a single unified schema with 27 core entities organised into six domain groups: Party & KYC, Account, Transaction & Screening, AML Operations (Alert/Case/STR), Graph & Analytics, and Batch & Governance.

### 4.1 Database Topology

| Database             | Engine                                     | Purpose                                                        | Data Includes                                                                                                                                      |
|----------------------|--------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>aml_core</b>      | PostgreSQL 16+ (3-node Patroni)            | Primary operational DB                                         | Parties, accounts, transactions, alerts, cases, STRs, screening results, detection scenarios, batch job tracking, KPI snapshots, graph nodes/edges |
| <b>aml_audit</b>     | PostgreSQL 16+ (append-only)               | Immutable operational audit store                              | All system-level audit events (audit_log table) with hash-chain verification                                                                       |
| <b>aml_forensic</b>  | PostgreSQL 16+ (WORM, physically separate) | Tamper-proof forensic ledger for all manual human interactions | Alert dispositions, case decisions, rule changes, risk overrides, SAR lifecycle, access logs, Merkle trees, external witness references            |
| <b>aml_graph</b>     | PostgreSQL 16+ + Apache AGE                | Graph exploration + community detection                        | graph_node, graph_edge, graph_community, graph_node_community, graph_analytics_run                                                                 |
| <b>aml_analytics</b> | TimescaleDB (PG extension)                 | Aggregation + time-series + behavioral baselines               | KPI snapshots, scenario KPIs, ML feature store, batch_behavioral_baseline, batch_peer_group                                                        |
| <b>redis_cache</b>   | Redis 7.2+                                 | Low-latency cache                                              | Wallet risk scores, sanctions watchlist index, customer risk tiers, rate limit counters, session state                                             |
| <b>minio_store</b>   | MinIO (4-node erasure)                     | Object / evidence storage                                      | SAR PDFs, case evidence, ML model artifacts, daily batch Parquet datasets,                                                                         |

|  |  |  |                                            |
|--|--|--|--------------------------------------------|
|  |  |  | daily briefing PDFs,<br>historical exports |
|--|--|--|--------------------------------------------|

## 4.2 Canonical Transaction Schema

All transactions normalised into this canonical schema before processing:

| Field                         | Type              | Source           | Description                                                       |
|-------------------------------|-------------------|------------------|-------------------------------------------------------------------|
| <b>txn_id</b>                 | UUID v7           | System           | Internal transaction identifier                                   |
| <b>txn_reference</b>          | varchar(255)      | Original system  | External reference: SWIFT UETR, blockchain tx hash, FPS reference |
| <b>txn_rail</b>               | enum              | Ingestion        | FIAT   STABLECOIN                                                 |
| <b>txn_channel</b>            | enum              | Ingestion        | SWIFT   CHATS   FPS   JETCO   ACH   BLOCKCHAIN   SVF   CARD       |
| <b>txn_type</b>               | enum              | Ingestion        | CREDIT   DEBIT   TRANSFER   MINT   REDEEM   SWAP   BURN           |
| <b>originator_account_id</b>  | UUID (FK account) | Payment system   | Originator account reference                                      |
| <b>beneficiary_account_id</b> | UUID (FK account) | Payment system   | Beneficiary account reference                                     |
| <b>originator_party_id</b>    | UUID (FK party)   | Customer profile | Originator party (indirect via account)                           |
| <b>beneficiary_party_id</b>   | UUID (FK party)   | Customer profile | Beneficiary party (indirect via account)                          |
| <b>txn_amount</b>             | numeric(24,8)     | Original system  | Transaction amount in original currency                           |
| <b>txn_currency</b>           | char(3)           | Original system  | HKD   USD   USDC   USDT   HKDG                                    |
| <b>hkd_equivalent_amount</b>  | numeric(24,2)     | FX conversion    | HKD-equivalent for regulatory threshold comparisons               |
| <b>txn_status</b>             | enum              | System           | PENDING   COMPLETED   FAILED   HELD   BLOCKED                     |
| <b>initiated_at</b>           | timestampz        | Original system  | When transaction was                                              |

|                            |              |                |                                                                    |
|----------------------------|--------------|----------------|--------------------------------------------------------------------|
|                            |              |                | initiated                                                          |
| <b>settled_at</b>          | timestampz   | Settlement     | When transaction was settled / confirmed on-chain                  |
| <b>originator_country</b>  | char(2)      | KYC / wallet   | ISO 3166-1 originator jurisdiction                                 |
| <b>beneficiary_country</b> | char(2)      | Counterparty   | ISO 3166-1 beneficiary jurisdiction                                |
| <b>is_cross_border</b>     | boolean      | System         | True if originator and beneficiary jurisdictions differ            |
| <b>purpose_code</b>        | varchar(255) | Payment / user | Purpose of transfer code                                           |
| <b>processing_mode</b>     | enum         | System         | REALTIME   BATCH (indicates which path processed this transaction) |
| <b>ingestion_timestamp</b> | timestampz   | System         | When ingestion-service received the transaction                    |
| <b>is_aml_processed</b>    | boolean      | System         | True after TME (real-time) or batch scanner (T+1) has processed    |

Blockchain-specific transactions extend the canonical schema via the `stablecoin_txn_detail` table (Section 4.3.3), capturing on-chain metadata: transaction hash, smart contract address, token standard, blockchain analytics risk scores, and flags.

### 4.3 Converged Data Model — Entity Reference

The complete data model comprises 27 core entities. All entities live in the `aml_core` schema unless otherwise noted. FK relationships are validated at the application layer; referential integrity constraints are applied selectively to avoid cascading issues in event-driven ingestion.

#### 4.3.1 Party & KYC Domain

| Entity       | Key Columns                                                                                                                                                                                                                                                                                                                              | Purpose                                                                                                                                    |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>party</b> | <code>party_id</code> (UUID PK), <code>party_type</code> (INDIVIDUAL CORPORATE FI TRUST), <code>full_name</code> , <code>hkid_passport_no</code> , <code>date_of_birth</code> , <code>nationality_code</code> (ISO 3166-1), <code>country_of_residence</code> , <code>registered_address</code> , <code>customer_risk_rating</code> (LOW | Core party record. Every customer (individual, corporate, FI, trust) has exactly one party row. Risk rating drives tier-based rule gating. |

|                           |                                                                                                                                                                                                                                                       |                                                                                                                                |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|                           | MEDIUM HIGH PEP <br>SANCTIONED), is_pep,<br>is_sanctioned, sanctions_list_ref,<br>is_beneficial_owner_identified,<br>onboarding_channel<br>(FACE_TO_FACE REMOTE <br>iAM_SMART),<br>cdd_last_reviewed_at,<br>edd_required_at, is_active,<br>created_at |                                                                                                                                |
| <b>party_document</b>     | doc_id (UUID PK), party_id (FK),<br>doc_type (HKID PASSPORT BR CI <br>POA), doc_number, expiry_date,<br>issuing_country, is_verified,<br>verified_at                                                                                                  | Identity documents belonging to a party. Supports multiple document types per party.                                           |
| <b>party_relationship</b> | rel_id (UUID PK), party_id_from<br>(FK), party_id_to (FK),<br>relationship_type (DIRECTOR <br>SHAREHOLDER <br>BENEFICIAL_OWNER <br>GUARANTOR),<br>ownership_percentage,<br>effective_from, effective_to                                               | UBO and control relationships between parties. Used by batch structuring detection to traverse beneficial ownership chains.    |
| <b>watchlist_entry</b>    | watchlist_id (UUID PK), list_type<br>(OFAC_SDN UNSC EU HMT <br>HKMA_SANCTIONED PEP),<br>listed_name, alias_names (text),<br>entity_type, listing_date,<br>delisting_date, listing_reason,<br>last_synced_at                                           | Sanctions and PEP watchlist master data. Synced daily from external list providers. Drives both real-time and batch screening. |

### 4.3.2 Account Domain

| Entity         | Key Columns                                                                                                                                                                                                                                                                            | Purpose                                                                                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>account</b> | account_id (UUID PK),<br>account_type (CURRENT SAVINGS <br>SVF STABLECOIN_WALLET <br>CRYPTO_WALLET), account_rail<br>(FIAT STABLECOIN),<br>account_number, currency_code,<br>institution_code (BIC LEI),<br>account_status (ACTIVE <br>DORMANT FROZEN CLOSED),<br>account_risk_rating, | Unified account entity spanning fiat bank accounts and blockchain wallets. A single party may have multiple accounts across rails. Wallet custody type drives unhosted wallet EDD rules. |

|                           |                                                                                                                                                                                     |                                                                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | current_balance,<br>blockchain_address,<br>blockchain_network (ETH TRON BNB POLYGON), is_hosted_wallet,<br>wallet_custody_type (HOSTED UNHOSTED EXCHANGE),<br>opened_at, created_at |                                                                                                                                                 |
| <b>account_party_link</b> | link_id (UUID PK), account_id (FK),<br>party_id (FK), role (OWNER JOINT_HOLDER AUTHORISED_SIGNATORY BENEFICIARY), effective_from,<br>effective_to                                   | M:N link between parties and accounts with role qualifier.<br>Enables multi-account structuring detection across accounts with different roles. |

#### 4.3.3 Transaction & Screening Domain

| Entity                       | Key Columns                                                                                                                                                                                                                                                                                                                                                                                                                                     | Purpose                                                                                                                                                                       |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>transaction</b>           | txn_id (UUID PK), txn_reference,<br>txn_rail, txn_channel, txn_type,<br>originator_account_id (FK),<br>beneficiary_account_id (FK),<br>originator_party_id (FK),<br>beneficiary_party_id (FK),<br>txn_amount, txn_currency,<br>hkd_equivalent_amount,<br>txn_status, initiated_at, settled_at,<br>originator_country,<br>beneficiary_country,<br>is_cross_border, purpose_code,<br>processing_mode,<br>ingestion_timestamp,<br>is_aml_processed | Canonical transaction record for both fiat and stablecoin transfers.<br>processing_mode indicates whether the transaction was processed by real-time TME, T+1 batch, or both. |
| <b>stablecoin_txn_detail</b> | sc_txn_id (UUID PK), txn_id (FK),<br>blockchain_tx_hash,<br>smart_contract_address,<br>originator_wallet_address,<br>beneficiary_wallet_address,<br>block_number, token_standard (ERC-20 TRC-20 BEP-20),<br>token_amount, token_symbol (USDT USDC HKDG TCHN),<br>is_unhosted_wallet_involved,<br>ba_risk_score (0-100), ba_provider (CHAINALYSIS ELLIPTIC TRM),                                                                                 | On-chain metadata extending the canonical transaction for stablecoin transfers. ba_risk_score and ba_risk_flags sourced from blockchain analytics vendor API.                 |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                             |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | ba_risk_flags (MIXER DARKNET <br>RANSOMWARE SANCTIONS),<br>on_chain_timestamp                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                             |
| <b>screening_result</b>   | screening_id (UUID PK), txn_id (FK),<br>party_id (FK), screen_type<br>(SANCTIONS PEP <br>ADVERSE_MEDIA <br>BLOCKCHAIN_RISK), screen_mode<br>(REALTIME BATCH), match_status<br>(NO_MATCH POTENTIAL_MATCH <br>CONFIRMED_MATCH <br>FALSE_POSITIVE), match_score,<br>matched_watchlist_ref,<br>matched_name, screened_at,<br>review_decision (CLEARED <br>ESCALATED BLOCKED),<br>reviewed_at                                   | Records every screening event —<br>both real-time and batch — against<br>sanctions lists, PEP databases, and<br>adverse media. Links to both<br>transaction and party. T+1 batch<br>re-screening uses<br>screen_mode=BATCH. |
| <b>travel_rule_record</b> | travel_rule_id (UUID PK), txn_id<br>(FK), compliance_status<br>(COMPLIANT MISSING_DATA <br>PENDING FAILED),<br>originator_name,<br>originator_account_ref,<br>originator_address,<br>originator_cid_number,<br>beneficiary_name,<br>beneficiary_account_ref,<br>beneficiary_institution_lei,<br>above_hkd8000_threshold,<br>data_transmission_protocol<br>(IVMS101 TRISA OPENVASP),<br>transmitted_at, missing_fields_list | Travel Rule compliance record per<br>qualifying transaction (>= HKD<br>8,000). Captures full IVMS101 data.<br>Temporal workflow writes status<br>updates.                                                                   |

#### 4.3.4 AML Operations Domain — Detection, Alert, Case, STR

| Entity                    | Key Columns                                                                                                                                                                                                                                            | Purpose                                                                                                                                                                                                                                                       |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>detection_scenario</b> | scenario_id (UUID PK),<br>scenario_code, scenario_name,<br>scenario_category (STRUCTURING <br>LAYERING PLACEMENT <br>TRAVEL_RULE_BREACH <br>BLOCKCHAIN_RISK), applicable_rail<br>(FIAT STABLECOIN BOTH),<br>processing_mode (REALTIME <br>BATCH BOTH), | Defines every detection scenario —<br>both real-time rules and batch<br>analysis types. Parameterised<br>thresholds stored as JSONB for hot-<br>reload. Regulatory reference maps<br>each scenario to the specific<br>regulatory requirement it<br>addresses. |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                   |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | scenario_logic_description (text), threshold_config (jsonb), lookback_window_days, severity (CRITICAL HIGH MEDIUM LOW), status (ACTIVE INACTIVE TESTING), regulatory_reference (AMLO HKMA_GUIDELINE FATF_REC), effective_from                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                   |
| <b>aml_alert</b>              | alert_id (UUID PK), scenario_id (FK), primary_txn_id (FK), primary_party_id (FK), alert_type (TRANSACTION CUSTOMER NETWORK SCREENING TRAVEL_RULE), alert_mode (REALTIME BATCH), alert_status (OPEN UNDER_REVIEW CLOSED_TP CLOSED_FP ESCALATED), alert_priority (CRITICAL HIGH MEDIUM LOW), risk_score (0-100), alert_description (text), triggered_rule_detail (jsonb), flagged_amount_hkd, alert_generated_at, assigned_analyst_id (FK aml_user), sla_due_at, disposition (TRUE_POSITIVE FALSE_POSITIVE INCONCLUSIVE), closed_at | Core alert entity. alert_mode distinguishes real-time alerts from T+1 batch alerts. triggered_rule_detail (JSONB) captures the full rule evaluation context including scores, thresholds, and contributing factors. Links to detection_scenario for traceability. |
| <b>alert_transaction_link</b> | link_id (UUID PK), alert_id (FK), txn_id (FK), link_reason                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | M:N link allowing one alert to reference multiple related transactions (e.g., structuring cluster).                                                                                                                                                               |
| <b>aml_case</b>               | case_id (UUID PK), case_reference (e.g. CASE-2026-HK-00123), primary_party_id (FK), case_type (INVESTIGATION SAR_PREPARATION STR_FILED EDD_REVIEW), case_status (OPEN UNDER_INVESTIGATION PENDING_STR STR_FILED CLOSED), case_priority (CRITICAL HIGH MEDIUM LOW), ml_typology (STRUCTURING LAYERING CRYPTO_MIXING                                                                                                                                                                                                                | Investigation case aggregating one or more related alerts. Tracks the full investigation lifecycle from opening through SAR filing. ml_typology classification feeds back into ML model training.                                                                 |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                        |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | SANCTIONS_EVASION),<br>case_summary (text),<br>total_flagged_amount_hkd,<br>assigned_analyst_id (FK),<br>assigned_supervisor_id (FK),<br>opened_at, sla_due_at, str_filed<br>(boolean), str_id (FK), created_at                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                        |
| <b>case_alert_link</b>   | link_id (UUID PK), case_id (FK),<br>alert_id (FK), linked_at                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | M:N link between cases and alerts.<br>One case may consolidate multiple<br>alerts; one alert may be re-<br>assigned to a different case.                                                                               |
| <b>case_party_link</b>   | link_id (UUID PK), case_id (FK),<br>party_id (FK), party_role_in_case<br>(SUBJECT ASSOCIATE <br>COUNTERPARTY FACILITATOR)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Tracks all parties involved in or<br>relevant to an investigation case.<br>Supports network graph expansion<br>from the case context.                                                                                  |
| <b>case_activity_log</b> | log_id (UUID PK), case_id (FK),<br>performed_by_user_id (FK<br>aml_user), activity_type<br>(ASSIGNED NOTE_ADDED <br>ESCALATED STATUS_CHANGED <br>STR_SUBMITTED),<br>activity_description (text),<br>performed_at                                                                                                                                                                                                                                                                                                                                                                                         | Immutable chronological log of all<br>analyst actions on a case. Used for<br>SLA tracking and audit. All entries<br>also published to forensic.events.                                                                 |
| <b>str_report</b>        | str_id (UUID PK), case_id (FK),<br>str_reference (JFIU STREAMS 2 ref),<br>str_status (DRAFT SUBMITTED <br>ACKNOWLEDGED <br>SUPPLEMENTAL),<br>submission_channel<br>(STREAMS2_WEBFORM <br>STREAMS2_XML),<br>reporting_institution_name,<br>reporting_officer_name,<br>subject_details (text),<br>suspicious_activity_description<br>(text), reasons_for_suspicion (text<br>— SAFE approach indicators),<br>applicable_ordinances (text —<br>Cap.405 Cap.455 Cap.575),<br>total_amount_reported_hkd,<br>tipping_off_risk_assessed<br>(boolean), activity_period_from<br>(date), activity_period_to (date), | Suspicious Transaction Report<br>record. Captures full JFIU STREAMS<br>2 content. Each STR version is<br>recorded in forensic.sar_lifecycle.<br>T+1 batch can pre-populate STR<br>drafts for high-confidence findings. |

|                             |                                                |                                                                |
|-----------------------------|------------------------------------------------|----------------------------------------------------------------|
|                             | submitted_at, jfiu_case_ref                    |                                                                |
| <b>str_transaction_link</b> | link_id (UUID PK), str_id (FK),<br>txn_id (FK) | M:N link between STR and the transactions cited in the report. |

#### 4.3.5 Graph & Analytics Domain

| Entity                      | Key Columns                                                                                                                                                                                                                                                                                  | Purpose                                                                                                                                                                                                              |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>graph_node</b>           | node_id (UUID PK), node_type (PARTY ACCOUNT TRANSACTION IP_ADDRESS DEVICE BLOCKCHAIN_ADDRESS), node_label, source_entity_id (FK to source table), source_table, risk_score (0-100), node_properties (jsonb), is_flagged, last_updated_at                                                     | Materialised graph node representing any entity type in the transaction graph. source_entity_id and source_table enable traversal back to operational data. node_properties (JSONB) stores type-specific attributes. |
| <b>graph_edge</b>           | edge_id (UUID PK), source_node_id (FK), target_node_id (FK), edge_type (SENT_TO RECEIVED_FROM OWNS_ACCOUNT CONTROLS SAME_IP BLOCKCHAIN_TX), edge_weight (numeric HKD or frequency), txn_count (aggregated edge count), total_amount_hkd, hop_count, edge_properties (jsonb), event_timestamp | Materialised graph edge. When multiple transactions exist between the same parties, they are aggregated into a single weighted edge. Used by batch graph algorithms (PageRank, Louvain, cycle detection).            |
| <b>graph_community</b>      | community_id (UUID PK), community_algorithm (LOUVAIN LABEL_PROPAGATION WEAKLY_CONNECTED), run_date, community_size, community_risk_score, total_txn_amount_hkd, is_suspicious, investigation_status (UNREVIEWED UNDER_REVIEW CLEARED ESCALATED), linked_case_id (FK aml_case), detected_at   | Detected community/cluster from graph analytics runs. Communities with anomalous transaction patterns are flagged for investigation. T+1 batch graph analysis updates community membership daily.                    |
| <b>graph_node_community</b> | gnc_id (UUID PK), node_id (FK), community_id (FK), membership_score                                                                                                                                                                                                                          | M:N link between nodes and communities with membership confidence score.                                                                                                                                             |
| <b>graph_analytics_run</b>  | run_id (UUID PK), algorithm_name                                                                                                                                                                                                                                                             | Tracks each execution of graph                                                                                                                                                                                       |

|  |                                                                                                                                                                                                                                                          |                                                                                           |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
|  | (PAGERANK BETWEENNESS LOUVAIN CYCLE_DETECTION SHORTEST_PATH), run_type (BATCH INTERACTIVE), run_date, run_parameters (text JSON), nodes_analysed, edges_analysed, suspicious_clusters_found, started_at, completed_at, status (RUNNING COMPLETED FAILED) | analytics — both automated T+1 batch runs and analyst-initiated interactive explorations. |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|

#### 4.3.6 Batch & Governance Domain

| Entity              | Key Columns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Purpose                                                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>batch_job</b>    | job_id (UUID PK), job_name, job_type (T1_TM_RUN T1_SCREENING T1_KPI_CALC T1_GRAPH_REFRESH T1_TRAVEL_RULE_CHECK), job_status (PENDING RUNNING COMPLETED FAILED), batch_date (processing date T), business_date (business date T-1), started_at, completed_at, txn_records_processed, alerts_generated                                                                                                                                                                                                         | Tracks each T+1 batch job execution. Maps to the Airflow DAG stages. batch_date is the processing date (when the scan runs); business_date is the data date (the previous day being scanned).  |
| <b>kpi_snapshot</b> | kpi_id (UUID PK), snapshot_date, kpi_period (DAILY WEEKLY MONTHLY QUARTERLY), applicable_rail (FIAT STABLECOIN ALL), total_active_customers, total_transactions_count, total_alerts_generated, true_positive_alerts, false_positive_alerts, false_positive_rate, total_strs_filed, str_conversion_rate, alert_to_str_rate, travel_rule_compliant_count, travel_rule_non_compliant_count, travel_rule_compliance_rate, alert_rate_per_1000_customers, avg_alert_review_hours, sla_breach_count, calculated_at | Pre-computed KPI snapshots for HKMA regulatory returns and internal dashboards. Includes all key HKMA-mandated KPIs: false positive rate, STR conversion rate, alert rate per 1,000 customers. |
| <b>scenario_kpi</b> | scenario_kpi_id (UUID PK),                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Per-scenario effectiveness metrics                                                                                                                                                             |

|                  |                                                                                                                                                                                                            |                                                                                                                                                                     |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | scenario_id (FK detection_scenario), snapshot_date, alerts_generated, true_positives, false_positives, false_positive_rate, strs_from_scenario, tuning_recommendation (INCREASE DECREASE RETIRE NO_CHANGE) | used by T+1 batch rule effectiveness retrospective analysis. Feeds tuning_recommendation back to compliance for periodic rule review.                               |
| <b>aml_user</b>  | user_id (UUID PK), full_name, email, role (ANALYST_L1 ANALYST_L2 SUPERVISOR MLRO GRAPH_ANALYST ADMIN), is_active, created_at                                                                               | Compliance team user registry. Synced from Keycloak. Role drives RBAC in AML Admin Portal. GRAPH_ANALYST role has elevated query access to graph exploration tools. |
| <b>audit_log</b> | audit_id (UUID PK), user_id (FK aml_user), table_name, record_id, action (INSERT UPDATE DELETE VIEW EXPORT), old_values (jsonb), new_values (jsonb), ip_address (inet), performed_at                       | System-level audit log capturing all data mutations and sensitive reads. Stored in aml_audit database (append-only, separate from aml_core).                        |

#### 4.3.7 Batch-Specific Tables (aml\_analytics / TimescaleDB)

| Entity                   | Key Columns                                                                                                                                                            | Purpose                                                                                                                           |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>batch_scan_run</b>    | run_id (UUID PK), scan_date, status (PENDING RUNNING COMPLETED FAILED), started_at, completed_at, txns_processed, alerts_generated, analysis_results (jsonb)           | Tracks each daily batch scan execution. analysis_results (JSONB) stores per-analysis breakdown (counts, durations, top findings). |
| <b>batch_scan_result</b> | result_id (UUID PK), run_id (FK), transaction_id, analysis_type, anomaly_score, findings (jsonb), alert_generated (bool), created_at                                   | Per-transaction batch analysis results, linked to batch run. One transaction may have multiple results (one per analysis type).   |
| <b>batch_alert</b>       | alert_id (UUID PK), run_id (FK), transaction_id, analysis_type, risk_score, alert_reason (text), priority (LOW MEDIUM HIGH), status, case_id (FK nullable), created_at | T+1 batch alerts. Lower default priority than real-time alerts. Links to aml_case when escalated for investigation.               |
| <b>batch_peer_group</b>  | group_id (UUID PK), group_name, segment, jurisdiction,                                                                                                                 | Peer group definitions and baseline statistics. Re-computed monthly.                                                              |

|                                  |                                                                                                                                                                      |                                                                                                                       |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
|                                  | income_band, product_usage, member_count, median_daily_volume, iqr_low, iqr_high, updated_at                                                                         | Used by T+1 batch peer group comparative analysis.                                                                    |
| <b>batch_behavioral_baseline</b> | customer_id (FK party), baseline_date, total_volume_30d, avg_txn_amount_30d, txn_count_30d, beneficiary_diversity_30d, rail_mix_ratio_30d, stddev_volume, updated_at | Per-customer 30/90-day behavioral baselines (TimescaleDB hypertable, 90-day chunks). Updated daily by T+1 batch scan. |

## 4.4 Entity Relationship Summary

The following summarises the key relationships between the 27 entities:

- party ||--o{ party\_document (a party has identity documents)
- party ||--o{ party\_relationship (a party relates to other parties — UBO, director, shareholder)
- party ||--o{ account\_party\_link (a party is linked to accounts with a role)
- account ||--o{ account\_party\_link (an account has one or more party links)
- account ||--o{ transaction (an account originates or receives transactions)
- party ||--o{ transaction (a party originates or receives transactions — denormalised for query performance)
- transaction ||--o{ stablecoin\_txn\_detail (stablecoin transactions have on-chain metadata)
- transaction ||--o{ travel\_rule\_record (transactions >= HKD 8,000 have Travel Rule records)
- transaction ||--o{ screening\_result (transactions are screened; parties are also screened independently)
- party ||--o{ screening\_result (parties are screened against watchlists)
- detection\_scenario ||--o{ aml\_alert (scenarios trigger alerts)
- detection\_scenario ||--o{ scenario\_kpi (scenario effectiveness is measured)
- aml\_alert ||--o{ alert\_transaction\_link (alerts link to one or more transactions)
- transaction ||--o{ alert\_transaction\_link (transactions are linked to alerts)
- aml\_alert ||--o{ case\_alert\_link (alerts are grouped into cases)
- aml\_case ||--o{ case\_alert\_link (cases contain alerts)
- aml\_case ||--o{ case\_party\_link (cases involve parties)
- party ||--o{ case\_party\_link (parties are subjects of cases)
- aml\_case ||--o{ case\_activity\_log (case activity is logged)
- aml\_case ||--o{ str\_report (cases may result in an STR filing)
- str\_report ||--o{ str\_transaction\_link (STRs reference transactions)
- transaction ||--o{ str\_transaction\_link (transactions are cited in STRs)
- graph\_node ||--o{ graph\_edge (nodes are sources and targets of edges)
- graph\_node ||--o{ graph\_node\_community (nodes belong to communities)
- graph\_community ||--o{ graph\_node\_community (communities contain nodes)

- aml\_case ||--o| graph\_community (suspicious communities may be linked to a case)
- aml\_alert ||--|| aml\_user (alerts are assigned to an analyst)
- aml\_case ||--|| aml\_user (cases are assigned to an analyst)
- audit\_log ||--|| aml\_user (audit entries reference the user who performed the action)

## 5. API Contracts & Service Interfaces

### 5.1 External REST API (AML Gateway)

| Endpoint                               | Method | Description                                                         |
|----------------------------------------|--------|---------------------------------------------------------------------|
| /api/v1/transactions/screen            | POST   | Submit txn for pre-settlement screening; sync HOLD/APPROVE response |
| /api/v1/transactions/batch-screen      | POST   | Async batch screening; returns batch_id for polling                 |
| /api/v1/transactions/batch/{id}/status | GET    | Poll batch screening status                                         |
| /api/v1/wallets/screen                 | POST   | Submit wallet address(es) for risk scoring                          |
| /api/v1/transactions/{ref}             | GET    | Retrieve transaction screening detail                               |
| /api/v1/health                         | GET    | Health check                                                        |
| /api/v1/forensic/verify                | GET    | Public hash-chain verification endpoint for auditors                |

### 5.2 Batch Scanning API (T+1)

| Endpoint                                 | Method | Description                                                                 |
|------------------------------------------|--------|-----------------------------------------------------------------------------|
| /api/v1/batch/scan/trigger               | POST   | Manually trigger a T+1 batch scan for a specific date; returns run_id       |
| /api/v1/batch/scan/{run_id}/status       | GET    | Poll batch scan progress: PENDING/RUNNING/COMPLETED/FAILED                  |
| /api/v1/batch/scan/{run_id}/results      | GET    | Retrieve full results for a completed batch scan: alerts, scores, anomalies |
| /api/v1/batch/scan/daily-briefing/{date} | GET    | Retrieve the Daily AML Briefing PDF/report for a given date                 |
| /api/v1/batch/scan/run-history           | GET    | List historical batch runs with status, duration, alert count               |
| /api/v1/batch/scan/peer-groups           | GET    | List peer group definitions and current baseline statistics                 |

## 5.2 Key Internal Service APIs

| Service                      | Interface      | Key Endpoints / Topics                                                                                                             |
|------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------|
| sanctions-service            | REST           | POST /screen; POST /screen/wallet                                                                                                  |
| wallet-analytics-adapter     | REST           | POST /score; GET /cache/status                                                                                                     |
| risk-scoring-service         | REST           | POST /score/transaction; GET /customer/{id}/risk-profile                                                                           |
| tme-engine                   | REST           | POST /decide; POST /decide/batch                                                                                                   |
| travelrule-gateway           | Temporal       | TravelRuleWorkflow(txn_payload)                                                                                                    |
| case-management-service      | REST           | GET /alerts; POST /cases; POST /cases/{id}/transitions                                                                             |
| regulatory-reporting-service | REST           | POST /sar/generate; POST /filing/submit                                                                                            |
| audit-service                | Kafka          | Consumes audit.events; no external API                                                                                             |
| forensic-audit-service       | Kafka + REST   | Consumes forensic.events; sole writer to ForensicDB; GET /forensic/records; GET /forensic/chain/verify                             |
| batch-scanning-service       | Airflow + REST | DAG triggered daily 02:00 HKT; GET /batch/scan/{run_id}/status; GET /batch/scan/daily-briefing/{date}; produces batch.alerts topic |

## 6. Infrastructure & Deployment (On-Prem / Hybrid)

### 6.1 On-Prem Kubernetes Cluster

| Component                 | Spec                                           | Sizing                                                                                        |
|---------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Kubernetes                | 3 CP + 6 workers (scale to 10)                 | k3s/RKE2 on Ubuntu 24.04;<br>64GB/16vCPU per worker                                           |
| PostgreSQL (aml_core)     | 3-node Patroni + 2 replicas                    | NVMe; 128GB/8vCPU; pgBackRest<br>to MinIO                                                     |
| PostgreSQL (aml_audit)    | 2-node HA (Patroni)                            | NVMe; 64GB/8vCPU; time-<br>partitioned tables                                                 |
| PostgreSQL (aml_forensic) | 2-node HA (Patroni) separate VLAN<br>with WORM | NVMe; 128GB/8vCPU; WORM<br>snapshots; S3 Object Lock backups;<br>annual WORM optical archival |
| Kafka/Redpanda            | 3-broker                                       | 32GB/8vCPU; 2TB NVMe per<br>broker; 7d retention                                              |
| Redis                     | 2-node active-passive                          | 32GB; wallet cache, sanctions<br>index, rate limiting                                         |
| MinIO                     | 4-node erasure coding                          | 8TB/node; evidence, model<br>artifacts, backups                                               |
| Blockchain nodes          | 1 per chain                                    | ETH archive: 2TB+; SOL: 1TB; L2s:<br>lighter RPC nodes                                        |
| Load balancers            | HAProxy + Nginx Ingress                        | Active-passive; TLS with HSM-<br>backed keys                                                  |
| HSM                       | Thales Luna / Azure Dedicated HSM              | Root CA, TLS, encryption, TRP<br>signing keys                                                 |
| Monitoring                | Prometheus + Grafana + Loki +<br>Tempo         | Metrics 30d; logs 90d; traces 14d                                                             |
| Backup                    | Velero + pgBackRest                            | Daily to MinIO; weekly off-site<br>encrypted cloud; WORM media<br>annual for forensic         |

### 6.2 Network Topology

- DMZ: API gateway + blockchain RPC ports (only external reachable components)
- Application: all microservices, Kafka, Redis (internal only; no direct external access)
- Data: PostgreSQL clusters, MinIO (isolated; accessed only via mTLS)
- Forensic Data: separate VLAN segment for ForensicDB; read-only replica for verifier/auditors; no write path externally
- Management: jumpbox with SSO + MFA; no direct SSH to nodes

- Segmentation: VLANs via Calico network policy; Kubernetes NetworkPolicies (default deny)

### 6.3 Kubernetes Resource Quotas

| Namespace         | Services                                                            | CPU                               | Memory          | Pods                                     |
|-------------------|---------------------------------------------------------------------|-----------------------------------|-----------------|------------------------------------------|
| <b>ingestion</b>  | ingestion-service,<br>blockchain-indexer                            | 4 cores                           | 16 GB           | 3/12                                     |
| <b>monitoring</b> | tme-engine, tme-<br>ml, sanctions-<br>service                       | 8 cores                           | 32 GB           | 3/16                                     |
| <b>analytics</b>  | wallet-analytics-<br>adapter, risk-<br>scoring-service              | 4 cores                           | 16 GB           | 2/10                                     |
| <b>compliance</b> | travelrule-gateway,<br>case-management,<br>regulatory-<br>reporting | 6 cores                           | 24 GB           | 3/10                                     |
| <b>forensic</b>   | forensic-audit-<br>service                                          | 2 cores                           | 8 GB            | 2/4                                      |
| <b>portal</b>     | aml-portal, api-<br>gateway                                         | 2 cores                           | 8 GB            | 2/6                                      |
| <b>infra</b>      | Kafka, Redis, PG<br>operators,<br>Temporal, Airflow                 | 12 cores                          | 48 GB           | Dedicated                                |
| <b>audit</b>      | audit-service                                                       | 2 cores                           | 8 GB            | 2/4                                      |
| <b>batch</b>      | batch-scanning-<br>service (Airflow-<br>triggered K8s Jobs)         | 16 cores (elastic,<br>job-scoped) | 64 GB (elastic) | 0/20 (jobs<br>create/destroy per<br>run) |

## 7. CI/CD Pipeline & DevSecOps

### 7.1 Pipeline Architecture

- Source: GitLab (self-hosted, on-prem) - single monorepo with per-module CI
- CI runner: GitLab Runner (Kubernetes executor, on-prem) - all builds in data centre
- Registry: GitLab Container Registry (on-prem, S3-compatible backing)
- Deploy: GitOps via ArgoCD (3-min sync; manual emergency; canary/blue-green per risk level)
- Secrets: Sealed Secrets for Git-encrypted; HashiCorp Vault for dynamic (DB creds, API keys); HSM for encryption/signing keys

### 7.2 Per-Service CI Pipeline Stages

- Lint: ruff/ESLint/gofmt
- Type check: mypy (strict) / tsc --noEmit
- Unit test: pytest (>85% coverage) / vitest / go test
- Security scan: Bandit, Trivy, Semgrep
- Build: Docker with distroless base images
- Push: GitLab Container Registry (Cosign signed)
- Integration test: ephemeral K8s namespace; service-level suite
- Deploy: ArgoCD PR-based sync to env-specific Git repo

## 8. Build Phases & Milestones (52-Week Roadmap)

### Sprint 0: Platform & Infrastructure Foundation (Weeks 0-4)

*Objective: Standing up dev and staging environments.*

- Provision on-prem K8s (dev + staging); deploy PostgreSQL, Kafka, Redis, MinIO, Temporal
- Deploy GitLab, Runner, ArgoCD, Harbor registry, Keycloak (RBAC roles)
- Set up Prometheus + Grafana + Loki + Tempo monitoring
- Establish CI/CD pipeline scaffolds for Python, TS, Go services
- Exit: developer merges to dev; deployed within 15 min; monitoring dashboards green

### Phase 1: Core Monitoring Engine (Weeks 4-14)

*Objective: Minimum viable AML - fiat + stablecoin through basic rule engine with sanctions.*

- Build ingestion-service + tme-engine v1 (configurable rules DSL, velocity/threshold/counterparty)
- Build sanctions-service v1 (OFAC SDN, exact + fuzzy matching); audit-service (append-only, hash-chain)
- Build blockchain-indexer (Ethereum only); integrate payment platform /screen endpoint
- Develop aml\_core PostgreSQL schema; document OpenAPI 3.1 contracts
- Exit: 100 fiat tps + 50 on-chain tps; sanctions <500ms p95; live on staging

### Phase 2: Multi-Chain & Wallet Analytics (Weeks 10-22)

*Objective: Full blockchain coverage with wallet risk scoring.*

- Extend indexer to Polygon, Arbitrum, Optimism, Base, Solana
- Build wallet-analytics-adapter v1 (abstracted vendor, Redis cache); risk-scoring-service v1
- Add wallet screening to sanctions; integrate real-time sanctions list updates
- Load testing: target 500 TPS
- Exit: all 6 chains within 30s of block finality; >100K txns/day processed

### Phase 3: Travel Rule & Case Management (Weeks 18-34)

*Objective: Full compliance - Travel Rule, analyst CMS, SAR reporting.*

- Build travelrule-gateway (Temporal TRP workflow, Notabene)
- Build case-management-service (state machine, investigation workspace, note/evidence); AML Admin Portal (Modules 1-3)
- Build regulatory-reporting-service v1 (JFIU-compatible SAR); deploy Airflow
- Exit: Travel Rule operational; CMS L1/L2/L3 swimlanes functional; portal feature-complete

### Phase 4: Intelligence & Regulatory Readiness (Weeks 30-44)

*Objective: ML scoring, graph, audit/forensic hardening, T+1 batch scanning for regulator inspection.*

- Deploy tme-ml inference (XGBoost ONNX, shadow mode); Airflow ML training pipeline

- Deploy aml\_graph (Apache AGE); ForensicDB and forensic-audit-service (tamper-proof manual interaction ledger)
- Build batch-scanning-service: Airflow DAGs for T+1 data extraction, enrichment, and batch analysis
- Implement batch-only analyses: full-day behavioral baselining, peer group comparison, multi-account structuring, network graph anomaly detection, ensemble ML scoring, cross-rail layering
- Deploy batch results storage in TimescaleDB; build Daily AML Briefing report generation
- PEP/adverse media screening; DR/BCP (hot standby); penetration test; CIS benchmark audit
- Regulatory documentation pack; SAR workflow end-to-end test with JFIU simulation
- External witness publishing activated (hourly blockchain + daily cloud attestation)
- Exit: pentest passed; forensic hash-chain verified; SAR tested; T+1 batch scanning producing daily briefings; documentation complete

## Phase 5: Optimisation & Scale (Weeks 40-52+)

*Objective: Production ML, expanded chains, performance.*

- Graduate ML to production (active scoring + rules); false positive reduction tuning
- Graduate batch ML to production (full feature ensemble, graph-based detection)
- Add Tron, Cosmos chain support; performance optimisation (queries, Kafka, Redis)
- Production rollout: phased traffic migration (10% > 50% > 100%); 30-day hypercare
- Exit: all milestones met; operating at target scale in production; regulatory filing live

### 8.1 Milestone Summary

| Phase           | Weeks  | Key Deliverables                                                                                                  |
|-----------------|--------|-------------------------------------------------------------------------------------------------------------------|
| <b>Sprint 0</b> | 0-4    | K8s + stateful services + CI/CD + monitoring live                                                                 |
| <b>Phase 1</b>  | 4-14   | TME v1, sanctions, Ethereum indexer, audit, fiat API                                                              |
| <b>Phase 2</b>  | 10-22  | All 6 chains, wallet analytics, risk scoring, live sanctions                                                      |
| <b>Phase 3</b>  | 18-34  | Travel Rule TRP, CMS, admin portal (dashboard/alerts/cases), SAR gen                                              |
| <b>Phase 4</b>  | 30-44  | ML (shadow), AGE graph, ForensicDB, T+1 batch scanning (full suite), PEP, DR/BCP, security audit, regulatory pack |
| <b>Phase 5</b>  | 40-52+ | ML production, false positive reduction, batch ML graduation, additional chains, production launch                |



## 9. Team Structure & Sizing

Four squads aligned to delivery phases. Peak build team: 20-25 engineers + supporting roles.

### 9.1 Squad Model

| Squad                                | Focus                                                                 | Active              | Headcount                      |
|--------------------------------------|-----------------------------------------------------------------------|---------------------|--------------------------------|
| <b>A: Core Platform</b>              | Ingestion, TME, sanctions, audit, batch enrichment integration        | Sprint 0 to Phase 5 | 5-6 (4 BE, 1-2 DE)             |
| <b>B: Blockchain &amp; Analytics</b> | Indexer, wallet analytics, ML, risk scoring, batch ML                 | Phase 1 to 5        | 4-5 (2 blockchain, 2 ML, 1 DE) |
| <b>C: Compliance UX</b>              | CMS, Travel Rule, portal, reporting, forensic, batch scanning service | Phase 2 to 5        | 5-6 (3 BE, 2 FE, 1 DE)         |
| <b>D: Platform &amp; Infra</b>       | K8s, CI/CD, security, monitoring, data platform, Airflow DAGs         | Sprint 0 to 5       | 4-5 (2 DevOps, 1 Sec, 1-2 DE)  |

### 9.2 Full Roles List

| Role                                  | Count | Key Skills                                                                                  |
|---------------------------------------|-------|---------------------------------------------------------------------------------------------|
| <b>Backend Engineer (Python)</b>      | 7     | FastAPI, async Python, Kafka, PostgreSQL, Temporal, Airflow, data model design              |
| <b>Backend Engineer (Go)</b>          | 2     | Go, blockchain nodes, gRPC, high-throughput                                                 |
| <b>Frontend Engineer (TS)</b>         | 2     | Next.js, React, Cytoscape.js, Recharts, shadcn/ui, graph visualisation                      |
| <b>Data Engineer</b>                  | 4     | Kafka, Airflow, PostgreSQL, ML pipelines, batch processing, Parquet, TimescaleDB, graph ETL |
| <b>Graph Analyst / Data Scientist</b> | 1     | Apache AGE, graph algorithms (PageRank, Louvain), network analysis, AML typology modelling  |
| <b>ML Engineer</b>                    | 2     | XGBoost, ONNX, feature engineering, SHAP, model lifecycle, batch inference                  |
| <b>DevOps / Platform Engineer</b>     | 3     | K8s (on-prem), Rancher, ArgoCD, GitLab CI, Terraform                                        |

|                          |   |                                                                                          |
|--------------------------|---|------------------------------------------------------------------------------------------|
| <b>Security Engineer</b> | 1 | CIS hardening, pentesting, HSM, PKI, secrets, network security                           |
| <b>QA / SDET</b>         | 3 | Chaos, k6 load, integration, regulatory scenario testing, batch pipeline testing         |
| <b>Technical Lead</b>    | 1 | System architecture, API contracts, data model governance, regulatory                    |
| <b>Delivery Manager</b>  | 1 | Vendor management, timeline, risk, stakeholder reporting                                 |
| <b>Compliance SME</b>    | 1 | Rule authoring, regulatory, SAR workflows, test scenarios, detection scenario definition |

## 10. Vendor Integration Matrix

| Vendor                 | Product              | Integration         | Phase              | Data Residency                           |
|------------------------|----------------------|---------------------|--------------------|------------------------------------------|
| TRM Labs               | Wallet Screening API | REST API            | Phase 2            | Vendor cloud (pseudonymised hashes only) |
| Chainalysis            | KYT                  | REST API (fallback) | Phase 2            | Vendor cloud (pseudonymised)             |
| Notabene               | Travel Rule Platform | REST + webhooks     | Phase 3            | Vendor cloud (PII encrypted in transit)  |
| LexisNexis / Refinitiv | World-Check          | SFTP + REST         | Phase 1            | On-prem (list data) + vendor cloud (PEP) |
| Onfido / Jumio         | KYC verification     | Webhook + REST      | Phase 1 (consumed) | Vendor cloud (PDPO agreement)            |
| OpenVASP / Sygna       | Travel Rule fallback | REST (future)       | Phase 3+           | Vendor cloud                             |
| JFIU                   | STREAMS 2 filing     | Web portal + SFTP   | Phase 4            | HK government (STR data)                 |

*All vendor contracts must include DPA with HK PDPO compliance, pseudonymisation requirements for blockchain analytics, SOC2 Type II, and right-to-audit clauses.*

## 11. Testing Strategy

### 11.1 Test Pyramid

| Layer              | Scope                                | Frequency                 | Tooling                              | Target                                                             |
|--------------------|--------------------------------------|---------------------------|--------------------------------------|--------------------------------------------------------------------|
| Unit               | Individual functions/services        | Every commit              | pytest / vitest / go test            | >85% coverage                                                      |
| Integration        | Service-to-service, Kafka, DB        | Every MR                  | pytest + testcontainers, temporalite | All critical paths                                                 |
| Contract           | API contract (CDC)                   | Every MR                  | Pact + OpenAPI validation            | All public endpoints                                               |
| E2E                | Full flow: ingestion to audit        | Nightly + pre-release     | Custom harness + k6                  | Top 20 scenarios                                                   |
| Regulatory         | HKMA/FATF typologies                 | Pre-release + quarterly   | Curated test cases                   | All documented scenarios                                           |
| T+1 Batch          | Batch scan full pipeline end-to-end  | Weekly + pre-release      | Airflow DAG test + batch job tests   | All 10 batch analyses produce correct output; idempotency verified |
| Chaos              | Node failure, partition, broker loss | Monthly                   | Chaos Mesh / Litmus                  | Survive single-node failure                                        |
| Forensic integrity | Hash-chain tamper simulation         | Pre-release + weekly auto | forensic-verifier CLI                | Detect all tampering; P0 alert on break                            |
| Load / Perf        | 2x peak volume                       | Pre-release + quarterly   | k6 + custom loadgen                  | 500 TPS; p95 <500ms                                                |
| Security           | SAST, container, deps                | Every commit + weekly     | Bandit, Trivy, Semgrep, ZAP          | Zero critical CVEs                                                 |

### 11.2 Key Regulatory Test Scenarios

| Scenario            | Description                                                  | Expected Outcome                      | Source           |
|---------------------|--------------------------------------------------------------|---------------------------------------|------------------|
| Structuring         | Multiple stablecoin txns just under HKD 8,000                | Alert generated; case created         | HKMA AML/CFT     |
| Cross-rail layering | Fiat deposit > stablecoin > unhosted wallet > secondary fiat | Cross-rail alert; enhanced monitoring | FATF 2025 report |

|                                              |                                                                                                                                         |                                                                                                                            |                          |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Sanctions hit</b>                         | Beneficiary wallet linked to OFAC SDN entity                                                                                            | Blocked; mandatory reporting                                                                                               | OFAC / UN                |
| <b>Travel Rule threshold</b>                 | USDC transfer HKD 10,000 to external VASP                                                                                               | TRP message sent; beneficiary data returned; released                                                                      | AMLO Sch.2 s.13A         |
| <b>Unhosted wallet high-value</b>            | USDC >HKD 100,000 to unhosted wallet                                                                                                    | Blocked; EDD required                                                                                                      | HKMA guidelines          |
| <b>PEP transaction</b>                       | PEP customer sends HKD 500,000 offshore                                                                                                 | Flagged; EDD; senior analyst                                                                                               | AMLO Sch.2 s.10          |
| <b>Velocity breach</b>                       | 50+ txns to 50 addresses in 1 hour                                                                                                      | Alert; account frozen                                                                                                      | FATF red flags           |
| <b>Forensic tamper event</b>                 | Direct DB modification attempt on ForensicDB                                                                                            | Trigger blocks write; tamper alert generated; hash chain break detected                                                    | AMLO record-keeping      |
| <b>T+1 batch: multi-account structuring</b>  | 3 linked accounts each send HKD 7,000 to different wallets; real-time passes all; T+1 batch detects coordinated pattern                 | T+1 batch alert generated; cross-account structuring rule triggered; case created with all 3 accounts linked               | HKMA AML/CFT             |
| <b>T+1 batch: peer group anomaly</b>         | Retail customer (declared income HKD 300K/year) processes HKD 2M in a single day through 20 small transfers                             | T+1 behavioral deviation score >3 sigma; peer group anomaly alert generated; case created for source-of-funds verification | FATF risk-based approach |
| <b>T+1 batch: sanctions list retroactive</b> | Mid-day sanctions list update adds new entity; 3 txns earlier that day involved that entity wallet; real-time screening used stale list | T+1 batch re-screen catches all 3 txns; retrospective alert generated; SAR pre-populated for analyst review                | OFAC 50% rule / HKMA     |

## 12. Security Architecture

### 12.1 Principles

- Least privilege: minimum permissions for every service and human operator
- Defence in depth: network segmentation, encryption at rest/transit, application controls
- Zero trust: no implicit trust between services; all inter-service via mTLS
- Separation of duties: no user can both configure rules and review cases; admin roles segregated

### 12.2 Controls

| Category          | Controls                                                                                                                                       |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Network           | VLAN segmentation (DMZ/app/data/forensic/batch/mgmt); NetworkPolicies default deny; WAF on API gateway                                         |
| Transport         | TLS 1.3 external; mTLS inter-service (Istio/Linkerd); HSM-backed certificates                                                                  |
| Data at rest      | AES-256 for PostgreSQL (TDE); MinIO SSE; WORM storage policy (S3 Object Lock) for ForensicDB backups; Parquet files encrypted at rest on MinIO |
| AuthN             | Keycloak OIDC (human); OAuth2 client credentials (M2M); audience-restricted tokens                                                             |
| AuthZ             | RBAC fine-grained per module; ABAC for case-level data; ForensicDB read segregated from write                                                  |
| Audit             | Immutable operational audit trail + cryptographically-sealed forensic ledger; hourly hash-chain verification; batch pipeline audit trail       |
| Secrets           | HashiCorp Vault dynamic secrets (DB creds daily rotation); no secrets in env vars; Sealed Secrets for Git                                      |
| App               | OWASP Top 10: CSRF tokens, XSS headers, rate limiting, Pydantic validation, parameterised queries                                              |
| Supply chain      | Cosign image signing; Syft SBOM; Trivy dependency scanning; distroless base images                                                             |
| Incident response | 24/7 on-call for AML decisions; alerts on auth failures, API anomalies, audit chain breaks, forensic tamper, batch scan failures               |

## 13. Performance & Scalability Requirements

### 13.1 SLIs / SLOs

| SLI                               | Target                         | Method                              |
|-----------------------------------|--------------------------------|-------------------------------------|
| Sync screening latency (p50)      | <200 ms                        | API endpoint histograms             |
| Sync screening latency (p95)      | <500 ms                        | API endpoint histograms             |
| Sync screening latency (p99)      | <1,500 ms                      | API endpoint histograms             |
| Throughput (sustained)            | 500 TPS                        | Request rate via API gateway        |
| On-chain ingestion latency        | <30s from block finality       | Block vs ingestion timestamps       |
| Travel Rule workflow              | <5 min to completion           | Temporal workflow metrics           |
| Alert generation latency          | <10s from txn receipt          | Txn vs alert timestamps             |
| Dashboard freshness               | <60s from event                | Event vs dashboard query timestamps |
| Forensic chain verification       | <5 min full chain              | forensic-verifier wall clock time   |
| T+1 batch scan completion         | <4 hours (02:00-06:00 HKT)     | Airflow DAG duration                |
| T+1 batch data extraction         | <30 min for 500K txns          | Stage-level DAG metrics             |
| T+1 batch analysis (per analysis) | <20 min (parallel jobs)        | K8s Job duration per analysis       |
| T+1 batch alert generation        | <10 min                        | DAG stage duration                  |
| Daily briefing generation         | <5 min                         | Report generation wall clock        |
| Availability (sync screening)     | 99.95%                         | Health check success / total        |
| Availability (async)              | 99.9%                          | Kafka consumer lag <10K messages    |
| Availability (batch scanning)     | 99.5% (daily window, not 24/7) | DAG success rate                    |

### 13.2 Scalability Strategy

- Horizontal: stateless services scale pods on Kafka lag + CPU
- Database: read replicas for dashboards; time partitioning for txns/audit; PgBouncer pooling
- Kafka: 2x peak partition sizing; rebalance planned during maintenance windows
- Cache: Redis cluster for wallet scores (90%+ hit rate target); in-memory sanctions index
- Blockchain: per-chain goroutine pools; one slow chain never blocks others
- Batch: elastic Kubernetes Jobs for each analysis type; 16 CPU/64GB burst per job; results on TimescaleDB hypertables for efficient time-range queries
- Batch storage: daily Parquet datasets on MinIO (90-day retention); compressed columnar storage for efficient analytical queries

## Appendix A: Technology Stack Reference

| Category             | Technology                                                     | Version    |
|----------------------|----------------------------------------------------------------|------------|
| Backend (Services)   | Python (FastAPI, Pydantic, SQLAlchemy, Kafka-Python)           | 3.12+      |
| Backend (Blockchain) | Go (go-ethereum, solana-go)                                    | 1.22+      |
| Workflows            | Temporal.io (Python SDK)                                       | 1.24+      |
| Batch Orchestration  | Apache Airflow (DAGs + K8s Jobs)                               | 2.10+      |
| Batch Compute        | Kubernetes Jobs (elastic, per-analysis)                        | 1.30+      |
| Batch Data Format    | Apache Parquet (MinIO storage)                                 | Latest     |
| Frontend             | Next.js 14+, React 18+, TypeScript, Tailwind CSS 5+, shadcn/ui | Latest     |
| Viz                  | Cytoscape.js 3.30+, Recharts 2.12+                             | Latest     |
| DB Primary           | PostgreSQL 16+ with Patroni (HA)                               | 16+        |
| DB Time-series       | TimescaleDB (PG extension)                                     | 2.17+      |
| DB Graph             | Apache AGE (PG extension)                                      | 1.5+       |
| DB Forensic          | PostgreSQL 16+ WORM (separate instance)                        | 16+        |
| Streaming            | Apache Kafka / Redpanda                                        | 3.7+ / 24+ |
| Cache                | Redis                                                          | 7.2+       |
| Object Storage       | MinIO                                                          | 2024+      |
| Orchestration        | Kubernetes (k3s / RKE2)                                        | 1.30+      |
| CI/CD                | GitLab (self-hosted) + ArgoCD                                  | 17+        |
| Workflow Scheduler   | Apache Airflow                                                 | 2.10+      |
| Auth / IAM           | Keycloak                                                       | 25+        |
| Monitoring           | Prometheus, Grafana, Loki, Tempo                               | Latest     |
| Secrets              | HashiCorp Vault                                                | 1.18+      |
| HSM                  | Thales Luna / Azure Dedicated HSM                              | Commercial |

## Appendix B: Glossary of Terms

**AGE:** Apache Graph Extension - PG extension for graph database via openCypher

**AMLO:** Anti-Money Laundering Ordinance (Cap. 615), Hong Kong

**AnchorPoint:** HKDR stablecoin issued by SCB x HKT x Animoca Brands JV

**ArgoCD:** GitOps CD tool for Kubernetes

**Batch Scanning (T+1):** Next-day bulk analysis of the complete previous day's transaction set, complementing real-time monitoring with computationally intensive analyses: full-day behavioral baselining, peer group comparison, multi-account structuring detection, network graph anomaly detection, ensemble ML scoring, and cross-rail layering

**CDD:** Customer Due Diligence - verifying identity and assessing risk

**CMS:** Case Management System - AML alert investigation and SAR filing workflow

**Detection Scenario:** A configurable AML detection rule or batch analysis definition stored in the detection\_scenario table with parameterised thresholds (JSONB), regulatory references, and processing mode (REALTIME|BATCH|BOTH)

**EDD:** Enhanced Due Diligence - deeper KYC for high-risk customers

**FATF:** Financial Action Task Force - international AML/CFT standard-setter

**ForensicDB:** Tamper-proof WORM ledger with cryptographic sealing for all manual human interactions

**GENIUS Act:** US federal legislation (mid-2025) regulating stablecoin issuers under BSA

**GitOps:** Git as single source of truth for infrastructure/application config

**Graph Analytics:** Computational analysis of the transaction network graph using algorithms such as PageRank (influence scoring), Louvain community detection (cluster identification), betweenness centrality, and cycle detection. Tracks execution metadata in graph\_analytics\_run and detected communities in graph\_community

**HKMA:** Hong Kong Monetary Authority - banking and stablecoin regulator

**HSM:** Hardware Security Module - tamper-resistant cryptographic key management

**JFIU:** Joint Financial Intelligence Unit (HK) - receives STRs

**KPI Snapshot:** Pre-computed regulatory KPI metric stored in kpi\_snapshot for HKMA returns: false positive rate, STR conversion rate, alert rate per 1,000 customers, Travel Rule compliance rate

**KYC/KYB:** Know Your Customer / Know Your Business - identity and due diligence

**Kustomize:** Kubernetes configuration customisation tool

**Louvain Community Detection:** Graph algorithm identifying densely connected clusters of nodes (accounts/parties/wallets). Communities with anomalous transaction patterns are flagged for investigation in the graph\_community table

**mTLS:** Mutual TLS - both client and server present certificates

**MiCA:** Markets in Crypto-Assets Regulation (EU)

**MLRO:** Money Laundering Reporting Officer - designated compliance officer

**PageRank:** Graph algorithm measuring node influence by analysing the quantity and quality of incoming edges. Applied to the transaction graph to identify highly central parties/wallets in fund flow networks

**PDPO:** Personal Data (Privacy) Ordinance (Cap. 486), Hong Kong

**PEP:** Politically Exposed Person - subject to enhanced due diligence

**Peer Group Analysis:** T+1 batch technique that compares a customer's daily activity against statistical norms of customers with similar profiles (segment, jurisdiction, income, product usage)

**RKE2:** Rancher Kubernetes Engine 2 - enterprise K8s distribution

**SAR/STR:** Suspicious Activity/Transaction Report - filed with FIU

**Scenario KPI:** Per-detection-scenario effectiveness metrics stored in scenario\_kpi. Computed by T+1 batch rule effectiveness retrospective. Includes tuning\_recommendation (INCREASE | DECREASE | RETIRE | NO\_CHANGE)

**SFC:** Securities and Futures Commission - HK securities and VASP regulator

**T+1:** The next business day after a transaction occurs. T+1 batch scanning runs at 02:00 HKT daily, processing the complete previous day's transaction set

**Temporal:** Durable execution platform for workflow orchestration

**TME:** Transaction Monitoring Engine - core real-time rule evaluation and scoring engine

**TRP:** Travel Rule Protocol - secure exchange of originator/beneficiary data between VASPs

**Travel Rule:** FATF Recommendation 16 - require originator/beneficiary info with transfers

**UBO:** Ultimate Beneficial Owner - natural person controlling a legal entity

**VASP:** Virtual Asset Service Provider - virtual asset transfer, custody, or exchange services

**Watchlist Entry:** Sanctions list or PEP database record stored in watchlist\_entry. Sources include OFAC SDN, UNSC, EU, HMT, HKMA. Synced daily. Drives both real-time and batch screening

**WORM:** Write Once Read Many - storage that prevents modification or deletion after write